UiA Universitetet i Agder

# MAS508-G
# Lab 3 – System modeling and linearization Solution



# Tasks

- **Find the equilibrium point(s).**
- **Linearize the system about the specified equilibrium.**
- **Define the given input/output and derive the transfer function of the linearized model.**
- **Discuss stability from the transfer function poles.**

# Task 1: Pendulum with Nonlinear Friction (torque input)

A pendulum of length $L$, mass $m$, with damping proportional to $\sin(\dot{\theta})$ is driven by an external torque $\tau(t)$:

$$mL^2\ddot{\theta} + b\,\sin(\dot{\theta}) + mgL\sin(\theta) = \tau(t).$$

**I/O definition:** input $u(t) = \tau(t)$, output $y(t) = \theta(t)$.

1. Identify the equilibrium (use $\theta_e = 0$, $\dot{\theta}_e = 0$, $u_e = 0$).
2. Linearize about $(\theta_e, \dot{\theta}_e) = (0, 0)$.
3. Find $G_1(s) = \Theta(s)/\tau(s)$.
4. From the poles of $G_1(s)$, discuss stability.

# Task1: Solution

**System:**

$$mL^2\ddot{\theta} + b\,\sin(\dot{\theta}) + mgL\sin(\theta) = \tau(t), \quad y = \theta.$$

**Equilibrium:** $\theta_e = 0,\ \dot{\theta}_e = 0,\ \tau_e = 0.$

**Linearization about** $(0,0)$: $\sin(\theta) \approx \theta,\ \sin(\dot{\theta}) \approx \dot{\theta}.$

$$mL^2\ddot{\theta} + b\,\dot{\theta} + mgL\,\theta = \tau(t).$$

**Transfer function:** Taking Laplace (zero ICs),

$$(mL^2s^2 + bs + mgL)\,\Theta(s) = \tau(s) \quad \Rightarrow \quad G_1(s) = \frac{\Theta(s)}{\tau(s)} = \frac{1}{mL^2s^2 + bs + mgL}.$$

**Stability from poles:** Poles are the roots of $mL^2s^2+bs+mgL = 0$. For $m > 0$, $L > 0$, $g > 0$ and $b \geq 0$, coefficients are positive. The second-order Hurwitz conditions hold, so poles lie in the LHP (or on the imaginary axis only if $b = 0$). Hence:

- $b > 0$: *stable.*
- $b = 0$: poles at $\pm j\sqrt{g/L} \Rightarrow$ *marginally stable* (undamped oscillation).

### Equilibrium at $\theta = \pi$

Set $\tau(t) = 0$. The equilibrium is at

$$\theta^* = \pi, \quad \dot{\theta}^* = 0.$$

Let $\theta(t) = \pi + \phi(t)$ with small deviation $\phi(t)$. Using small-angle approximation

$$\sin(\theta) = \sin(\pi + \phi) = -\sin(\phi) \approx -\phi, \qquad \sin(\dot{\theta}) = \sin(\dot{\phi}) \approx \dot{\phi}.$$

Substitute into the dynamics:

$$mL^2\ddot{\phi} + b\dot{\phi} - mgL\phi = 0.$$

### Linearized System

Dividing by $mL^2$ gives

$$\ddot{\phi} + \frac{b}{mL^2}\dot{\phi} - \frac{g}{L}\phi = 0.$$

The characteristic polynomial is

$$\lambda^2 + \frac{b}{mL^2}\lambda - \frac{g}{L} = 0.$$

Thus the eigenvalues are

$$\lambda = -\frac{b}{2mL^2} \pm \sqrt{\left(\frac{b}{2mL^2}\right)^2 + \frac{g}{L}}.$$

## Stability Analysis

Since $\frac{g}{L} > 0$, the square root term is larger than $\frac{b}{2mL^2}$. Therefore, one eigenvalue is positive and one is negative. his implies that the equilibrium at $\theta = \pi$ is **unstable**, regardless of the value of $b > 0$.

## Task 2: Nonlinear Spring–Mass (force input)

A mass–spring with cubic stiffness is driven by a force $F(t)$:

$$m\ddot{x} + k_1 x + k_2 x^3 = F(t).$$

**I/O definition:** input $u(t) = F(t)$, output $y(t) = x(t)$.

1. Find equilibrium point(s). Consider linearization at $x_e = 0$, $u_e = 0$.
2. Linearize about $x_e = 0$.
3. Find $G_2(s) = X(s)/F(s)$.
4. Use the poles of $G_2(s)$ to assess stability.

# Task 2 : Solution

**System:**

$$m\ddot{x} + k_1 x + k_2 x^3 = F(t), \quad y = x.$$

**Equilibria:** $k_1 x_e + k_2 x_e^3 = F_e$. For linearization at the origin, take $x_e = 0,\ F_e = 0$.

**Linearization at** $x_e = 0$:

$$m\ddot{x} + k_1 x = F(t).$$

**Transfer function:**

$$(ms^2 + k_1)X(s) = F(s) \quad \Rightarrow \quad G_2(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + k_1}.$$

**Stability from poles:** Poles at $s = \pm j\sqrt{k_1/m}$ for $k_1 > 0$ (no damping given) $\Rightarrow$ *marginally stable*. If physical damping $c\dot{x}$ were present, denominator becomes $ms^2 + cs + k_1$ and the poles move to the LHP for $c > 0$.

# Task 3: Nonlinear Two-State System

$$\dot{x}_1 = -x_2 + x_1^2 + u, \qquad \dot{x}_2 = x_1 - x_2^3, \qquad y = x_1.$$

**I/O definition:** input $u(t)$ enters the $\dot{x}_1$ equation additively; output $y(t) = x_1(t)$.

1. Find all equilibrium points.
2. Linearize at each equilibrium to obtain $\dot{\delta x} = A\delta x + B\delta u,\ y = C\delta x$.
3. Derive the transfer function $G_3(s) = Y(s)/U(s) = C(sI - A)^{-1}B$.
4. Discuss stability from the poles of $G_3(s)$ at each equilibrium.

# Task 3: Solution

## System Definition

$$\begin{cases} \dot{x}_1 = -x_2 + x_1^2 + u, \\ \dot{x}_2 = x_1 - x_2^3, \\ y = x_1. \end{cases}$$

We seek to linearize via a first-order Taylor expansion about each equilibrium $(x_e, u_e)$. Recall the general formula (Taylor approx):

$$f(x,u) \approx f(x_e,u_e) + \left.\frac{\partial f}{\partial x}\right|_{(x_e,u_e)} (x - x_e) + \left.\frac{\partial f}{\partial u}\right|_{(x_e,u_e)} (u - u_e),$$

neglecting higher-order terms.

Let $\delta x = x - x_e$, $\delta u = u - u_e$. Then the linearized form is

$$\dot{\delta x} = A\,\delta x + B\,\delta u, \quad \delta y = C\,\delta x.$$

## Equilibrium Points

Setting $\dot{x}_1 = 0$ and $\dot{x}_2 = 0$ with $u_e = 0$:

$$-x_{2e} + x_{1e}^2 = 0, \qquad x_{1e} - x_{2e}^3 = 0.$$

Thus:

$$x_{2e} = x_{1e}^2, \quad x_{1e} - (x_{1e}^2)^3 = x_{1e}(1 - x_{1e}^5) = 0.$$

Solutions:

$$(x_{1e}, x_{2e}) = (0, 0) \quad \text{or} \quad (1, 1).$$

## Taylor Expansion and Obtaining $A$, $B$, $C$

Define the vector function:

$$f(x, u) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} -x_2 + x_1^2 + u \\ x_1 - x_2^3 \end{bmatrix}, \quad g(x) = y = x_1.$$

**Compute partial derivatives (Jacobian) based on Taylor expansion:**

$$\frac{\partial f_1}{\partial x_1} = 2x_1, \quad \frac{\partial f_1}{\partial x_2} = -1, \quad \frac{\partial f_1}{\partial u} = 1;$$

$$\frac{\partial f_2}{\partial x_1} = 1, \quad \frac{\partial f_2}{\partial x_2} = -3x_2^2, \quad \frac{\partial f_2}{\partial u} = 0;$$

$$\frac{\partial g}{\partial x} = [1 \quad 0], \quad \frac{\partial g}{\partial u} = 0.$$



symbolically,

$$A(x_e) = \begin{bmatrix} 2x_1 & -1 \\ 1 & -3x_2^2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

## At Equilibrium (0,0)

Substitute $(x_{1e}, x_{2e}) = (0, 0)$:

$$A_0 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad B,\ C \text{ as above.}$$

Thus the linearized model:

$$\dot{\delta x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \delta x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \delta u, \quad \delta y = [1\ 0]\delta x.$$

The transfer function:

$$G_{3,0}(s) = \frac{Y(s)}{U(s)} = \frac{s}{s^2 + 1}.$$

The poles are $s = \pm j$ on the imaginary axis—this yields **marginal stability** for the linearized model.

## At Equilibrium (1,1)

Substitute $(x_{1e}, x_{2e}) = (1, 1)$:

$$A_1 = \begin{bmatrix} 2 & -1 \\ 1 & -3 \end{bmatrix}.$$

Transfer function:

$$G_{3,1}(s) = \frac{s+3}{s^2+s-5}, \quad \text{poles are roots of } s^2+s-5=0,$$

i.e., $s = \dfrac{-1 \pm \sqrt{21}}{2}$. One pole in the Right Half Plane $\rightarrow$ **unstable ** behavior.